

(434MHz). This modulated signal is then transmitted through the antenna. The waveforms in Fig. 2 depict the ASK concept. The main specifications of the RF module are shown in the table.

Receiver section. Fig. 3 shows the block diagram of the receiver section.

The 12V DC supply, used along with a 5V regulator, can be provided by a 12V battery or power adaptor.

The RX-434 radio receiver module receives the ASK signal from TRX-434. The HT12D decoder demodulates the received address and data bits. IC CD4519 is a quadruple two-input multiplexer that selects the appropriate data bits to control the devices.

The ULN2003 relay driver consists of seven npn Darlington pairs that feature high-voltage outputs with common-cathode clamp diodes for

switching the inductive loads. The collector-current rating of a single Darlington pair is 500 mA.

Circuit description

Transmitter circuit. Fig. 4 shows the transmitter circuit. The microcontroller reads the input data from switches S1 through S4 at its port-2 pins 21 through 24 and displays it on the LCD. Port 3 provides read data to the encoder IC HT12E at pins 10 through 13. The microcontroller is programmed to control input and output data.

When the pushbutton switches (S1 through S4) are open, logic '0' is constantly fed to the respective port pins of the microcontroller. When any of the buttons is pressed, logic '1' is fed to the respective port pin of the microcontroller.

The device control program stored in the memory of the microcontroller activates and executes as per the functions defined in the program for

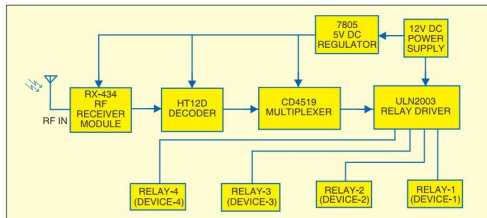


Fig. 3: Block diagram of receiver section for wireless equipment control

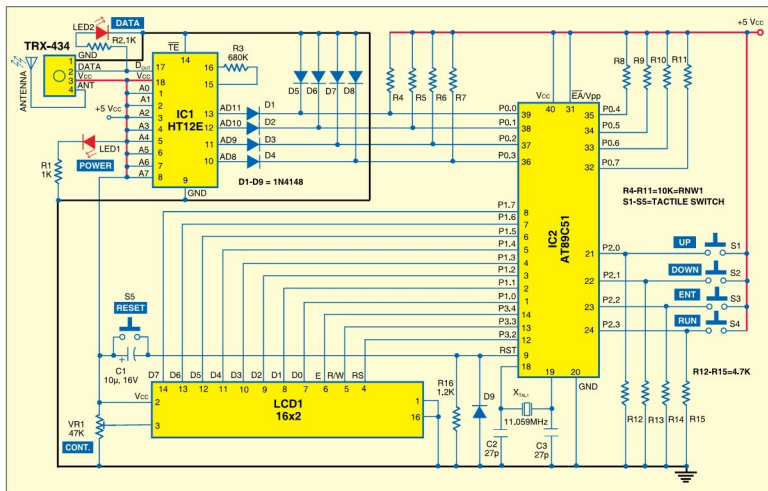


Fig. 4: Transmitter circuit